

Bootstrapping Multi-Modal Semantic-Vector Spaces

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We present an alignment and evaluation architecture for bootstrapping multimodal systems into a shared Semantic-Vector Space (SVS) (Fig. 1) that decouples encoding, inference, and decoding tasks so each module can be independently aligned and evaluated; our training pipeline leverages momentum-based cross-modal distillation as described in [1] to stabilize joint embedding learning across noisy cross-modal pairings.

To minimize compute and data costs, we align pretrained encoders and decoders using lightweight adapter MLPs[2](Fig. 2) and synthesize cross-modal training pairs via backtranslation and contrastive objectives, which was shown by [3] to be effective for generating high-fidelity synthetic pairings and organizing cross-modal latent structure.

We demonstrate a methodology for rapidly bootstrapping multimodal encoder and decoder swarms using open-weight vision, text, and audio models. This space can be used to further explore the utility of the Large Concept Model[4] and continuous-latent reasoning[5] approaches as well as exploring the applicability of statistical analysis and pattern recognition algorithms for exploring relationships in the semantic space.

Our results provide a reproducible methodology for aligning new modalities to a Shared SVS, highlights practical trade-offs between adapter-based alignment and full-model fine-tuning, and show that a shared semantic-vector modality can serve as an efficient, extensible foundation for robust multimodal AI.

References

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